

PAIN Relief: A Verified-Control Method for Reducing FedRAMP Rev5 VDR/VER PAIN and Exploitability

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Abstract

The companion memo *A Deterministic, CVSS–Environmental Method for FedRAMP Rev5 VDR/VER Vulnerability Prioritization* derives a finding’s Potential Agency Impact (PAIN) and its remediation deadline. That memo also establishes a mitigation *ladder*: PAIN comes down when cited, signed evidence lowers a Modified impact metric. This paper makes the ladder systematic. It defines **PAIN Relief**: a governed, versioned catalog that pairs a *machine-verified* security control with the specific weakness class (by CWE) it provably counters and the deterministic scoring reduction that pairing earns. Two levers move: verified controls that bound what a successful exploit can do lower a Modified impact metric and step the PAIN level down; verified controls that reduce the probability of successful exploitation lower a *local* estimate of that probability and can move a finding from likely-exploitable to not. Neither lever mutates a published input — not the CVSS base vector, not the EPSS score, not the CISA KEV clock. The method’s guiding line is *enforced-and-machine-verifiable*, not *claimed*: a control earns relief only when enforcement infrastructure applies it and a collector can read it. The catalog is offered as an example; each provider governs its own.

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1 Status and Relationship to the Companion Papers

This document is informational. It does not define a FedRAMP requirement; it specifies a method for lowering the PAIN and exploitability of a finding within the existing VDR/VER rules, building directly on the PAIN memo’s mitigation ladder. It consumes the internet-reachability determination of *An Exclusion-Gated Method for Determining Internet Reachability* and the disposition vocabulary of *A CycloneDX VEX Profile for FedRAMP Rev5 VDR/VER*. The key words MUST, SHOULD, and MAY follow RFC 2119.

2 The Problem: Good Hygiene Must Pay

A provider that runs its workloads with least privilege, deny-by-default egress, read-only filesystems, and enforced runtime protection is measurably harder to harm than one that does not. Yet a raw scan scores an identical CVE identically on both. If secure configuration changes no number a provider must report, it is all cost and no credit, and the rational provider treats hardening as a burden. The PAIN method already refuses that outcome for one input — asset criticality, through the CVSS Environmental requirements — by re-weighting a vulnerability’s impact by what is at stake on the specific asset. PAIN Relief extends the same principle from *what the asset is* to *what protects it*.

IN PLAIN TERMS

A scanner scores a flaw as if every system running it were equally exposed. Two providers run the same flaw; one has locked the system down and one has not. PAIN Relief is how the locked-down provider’s work shows up in the score — lower impact, or a longer clock — so that doing security well is worth something on paper, not just in principle.

2.1 The axis that matters: enforced, not claimed

The tempting mistake is to reward controls a provider *asserts*. Input sanitization, a WAF signature set, “we validate everything” — these are claims about behavior, unverifiable at assertion time and silently perishable on the next refactor. PAIN Relief admits a control only when it is **machine-verifiable** — read from enforced configuration or telemetry a collector can inspect (a NetworkPolicy, a `securityContext`, an RDS flag) — or carried by a signed, dated, reusable attestation for structure a platform cannot expose. A control whose effectiveness is a claim stays in the VEX disposition lane of the companion profile, decided per finding by a person. The line is not *a priori* versus *a posteriori*; it is *enforced-and-verifiable* versus *asserted*.

3 Two Levers

PAIN Relief moves exactly two things, and keeps them strictly apart. A control belongs to one lever, chosen by its causal mechanism: a control that bounds *what a successful exploit yields* moves impact; a control that reduces *whether the exploit succeeds* moves exploitability. Because the assignment follows the mechanism, a control never scores on both axes, and nothing double-counts.

3.1 Impact: lowering a Modified metric

The PAIN memo scores a finding through the CVSS Modified Impact Sub-Score, built from the per-dimension impacts C, I, A and the asset’s requirements. A verified control that provably counters a weakness class lowers the affected Modified impact metric from High to Low before that score is computed:

$$\text{High} \longrightarrow \text{Low}, \quad \text{never None.} \quad (1)$$

The floor is deliberate. A claim of *zero* impact is an applicability claim, and it travels as a VEX disposition (`code_not_present`, `code_not_reachable`), not as relief. The reduced C, I , or A flows through the memo’s existing arithmetic unchanged; the PAIN level and deadline follow. There is one rung, and it does not stack: several controls that each argue a dimension down still lower it by a single step, with every contributing control named in the finding’s evidence.

3.2 Exploitability: lowering a local estimate

FedRAMP’s exploitability axis is binary: a finding is likely-exploitable (LEV) or not (NLEV), and the two select different remediation columns. The memo computes

$$\text{LEV} = \text{KEV} \vee (\text{EPSS} \geq \tau) \vee \text{floor}, \quad \tau = 0.70,$$

where the floor is the FRD-LEV rule that an automated unauthenticated internet exploit is likely-exploitable. PAIN Relief never edits the published EPSS score — a global prediction is not a provider’s to rewrite — just as the impact lever never edits the CVSS base vector. Instead a verified control lowers a *local* estimate of exploitation probability, and the binary threshold does the rest:

$$\text{adjustedEPSS} = \max\left(\text{EPSS} \times \prod_{r \in R} \rho_r, \text{EPSS} \times \phi\right), \quad (2)$$

where R is the set of applicable verified likelihood controls, each carrying a *residual factor* $\rho_r \in (0, 1)$ — the share of exploitation risk it leaves behind — and ϕ is a governed floor on total reduction. LEV is then recomputed with `adjustedEPSS` in place of EPSS.

DEFINITION

ρ_r (*residual factor*): a control that halves exploitation likelihood carries $\rho = 0.5$; one that trims it modestly carries $\rho = 0.85$. Lower is stronger. ϕ (*stacking floor*): the most that stacking may reduce the estimate, e.g. $\phi = 0.5$ caps total reduction at one half. Both are governed, back-tested constants, documented and challengeable like the PAIN cut points.

Three properties of Equation (2) matter. First, **it stacks multiplicatively**: independent controls compound, so layered runtime defense is rewarded — this is the one place defense-in-depth earns more than any single control. Second, **the threshold self-calibrates**: a modest control drops a marginal finding ($0.72 \times 0.85 = 0.61$, now NLEV) but not a near-certain one ($0.99 \times 0.85 = 0.84$, still LEV), which is exactly right — one control should not defuse a vulnerability being exploited everywhere. Third, and absolutely, **KEV is frozen**: a Known Exploited Vulnerability stays LEV regardless of ρ , and its BOD 26-04 clock is untouched. The FRD-LEV floor is a reachability premise, not a probability, so it is defeated only in the binary — by a qualifying edge-authentication boundary — never by a residual factor.

IN PLAIN TERMS

EPSS is a worldwide guess at how likely a flaw is to be attacked. We never change that number. We keep a second, local number that starts at EPSS and drops when we can verify a control that makes the attack less likely to work here. If the local number falls below the line, the finding earns the slower clock. Stacked controls push it lower together — but a flaw that is already being exploited in the wild (KEV) never moves, and the change is always shown beside the original so nothing is hidden.

4 The Catalog

A PAIN Relief entry is a row pairing a control, the CWE class it counters, and the move it earns. Rows are keyed to CWE because the finding carries a CWE and the control’s causal story is a statement about a weakness class — “deny-by-default egress breaks the second-stage fetch of injection and staged-execution flaws” is a claim about **CWE-917**, **CWE-94**, and their kin, not about a single CVE. Where a control bounds what *any* code execution yields regardless of how it began, the row keys to an outcome class (all execution-capable weaknesses, including memory corruption) rather than an enumerated list. A finding with no CWE, or an unrecognized one, earns nothing: relief fails open toward the higher score.

Verified control	Counters (CWE)	Lever	Move
Egress default-deny	917, 94, 98, 829	impact	MC/MI High→Low
Read-only root filesystem	22, 434, 73	impact	MI High→Low
No shell in image	78, 77	impact	MC/MI High→Low
Least-privilege DB grant	89	impact	MC/MI High→Low
Verified HA + rate limit	crash-class DoS	impact	MA High→Low
Blocking-mode EDR/RASP	(any)	likelihood	$\rho = 0.85$
Edge auth (remote-access)	(any)	likelihood	defeats FRD-LEV floor

Table 1: Illustrative PAIN Relief rows. The maintained catalog is proprietary and provider-governed; this sample is an existence proof, not a standard — exactly the posture of the archetype catalog in the companion memo. Every applied credit ships its row identifier, rationale, and the catalog version in the finding’s evidence.

5 Governance

Adding a row grants a discount across the estate, so a row is a security setting and is reviewed like one. Each row **MUST** satisfy:

- **A causal story, not an adjective.** “Egress-deny breaks the staged fetch” qualifies; “EDR makes exploitation harder” does not. The row names the mechanism by which the control interrupts the weakness class.
- **Machine-verifiable or attested, never claimed.** The control is proven from enforced configuration, telemetry, or a signed reusable artifact. This is the clause that keeps input sanitization and WAF signature coverage *out* of the catalog and in the VEX lane: both are bets that a pattern matches every variant an adversary can invent, and Log4Shell’s same-week obfuscations are the standing proof they are not verifiable facts.
- **CWE-keyed and fail-open.** A row applies only to the classes it lists; missing or generic CWEs earn nothing.

- **One rung; never None.** Impact moves High to Low only; zero-impact is a VEX applicability claim. Exploitability moves the binary via Equation (2); nothing here touches reachability or KEV dates.
- **Versioned, pinned, and evidenced.** A score is reproducible only against a named catalog release; the row id and version travel in every evidence line. Refusals are recorded, so the catalog is curated, not accumulated.
- **Back-tested before adoption.** Residual factors, the stacking floor, and any new row are calibrated against real findings; a row that moves a large fraction of the estate is presumed miscalibrated until argued otherwise.

6 What PAIN Relief Never Does

No entry mutates the published EPSS score or the vendor CVSS base vector; both the original and the adjusted values appear in output. No entry alters the internet-reachability determination — that is the companion exclusion-gated method’s job, and PAIN Relief merely consumes its result. No entry extends or softens a CISA KEV / BOD 26-04 remediation date; a Known Exploited Vulnerability carries its clock untouched no matter how thoroughly it is mitigated. And no entry lowers a Modified impact dimension to None; that stronger claim is a signed VEX disposition, separately governed.

7 Worked Examples

Impact — egress-deny on a Log4Shell-class finding. A CWE-917 finding scores $I = \text{High}$ on an application asset, placing it at, say, N4. The workload runs under a verified default-deny egress NetworkPolicy with no wildcard allow. Log4Shell’s remote-code-execution requires an outbound fetch of a second-stage payload; the egress policy removes the outbound leg, so the injection still fires but cannot complete. The egress-deny row lowers Modified Integrity High→Low; recomputed through the memo’s arithmetic the finding lands at N3 — a longer deadline, earned by a control the plugin read directly from the cluster. The reachability answer is unchanged: the finding is still internet-reachable; what changed is what it can accomplish.

Exploitability — blocking EDR crossing the threshold. A finding carries EPSS 0.74 (LEV, since $0.74 \geq 0.70$), is not KEV, and the floor does not apply. The workload runs blocking-mode EDR in enforcing mode, verified from its policy export, carrying $\rho = 0.85$. Then $\text{adjustedEPSS} = 0.74 \times 0.85 = 0.63 < 0.70$, and the finding moves to NLEV — the slower remediation column. Had EPSS been 0.99, the same control would leave it at 0.84, still LEV: the threshold ensures a single control relieves the marginal case, not the near-certain one. Published EPSS remains 0.74 in the record beside the adjusted value.

IN PLAIN TERMS

Same two flaws, two providers. On the hardened estate the first drops a PAIN level and the second drops onto a slower clock, purely because a scanner-adjacent tool could *see* the controls that make those flaws less dangerous here. On the unhardened estate, nothing moves. That difference is the entire point.

8 Architecture: Capture and Evaluation Are Separate

The determination splits across a clean boundary. A collector with cluster access — the Kubernetes plugin — *captures* only: the finding’s CWEs and a neutral block of raw Kubernetes-observed facts per workload (`securityContext` settings, NetworkPolicy rules, replica and disruption posture, service-account and volume facts). It makes no interpretation: it emits that a NetworkPolicy denies egress to a link-local address, not that “IMDS is protected.” A downstream evaluator, which need not touch the cluster, loads the governed catalog, maps the captured facts onto the control predicates, and applies Equations (1) and (2). The split keeps the privileged collector simple and auditable and lets the catalog — the part that embodies judgment — evolve under governed review without redeploying the collector.

9 Conclusion

PAIN Relief closes the loop the PAIN memo opened. The memo re-weights a vulnerability by what is at stake on the asset; this paper re-weights it by the controls that verifiably protect the asset, using the same Environmental machinery and the same discipline of citing evidence. Impact steps down a rung when a verified control bounds the damage; exploitability crosses a threshold when verified controls make success less likely; and neither ever edits a published input, softens a KEV clock, or turns a claim into a discount. The result gives a provider what raw scanning never has: a deterministic, auditable reason that good security configuration lowers the number — and therefore a reason to build securely in the first place.